Fiscal Year:	FY 2020	**Task Last Updated:**	FY 12/06/2019
PI Name:	Baker, John Ph.D.		
Project Title:	Defining the Relationship Between Simulated Weightlessness and Space Radiation on Cardiovascular Disease, and the Effectiveness of a Countermeasure		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jbaker@mcw.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	414-955-8706
Organization Name:	Medical College of Wisconsin		
PI Address 1:	Congenital Heart Surgery		
PI Address 2:	8701 W Watertown Plank Rd		
PI Web Page:			
City:	Milwaukee	**State:**	WI
Zip Code:	53226-3548	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-SRHHC. Appendix E: Space Radiobiology and Human Health Countermeasures Topics
Start Date:	02/04/2019	**End Date:**	02/03/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	December 2019 report: None.		
COI Name (Institution):	Kronenberg, Amy Sc.D. (Lawrence Berkeley National Laboratory) Willey, Jeffrey Ph.D. (Wake Forest University)		
Grant/Contract No.:	80NSSC19K0498		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal is one of two proposals selected for funding in response to solicitation NNJ16ZSA001N-SRHHC “Topic 6: Combined Effects of Simulated Weightlessness and Space Radiation on Cardiovascular Structure and Function,” and then descoped and merged at the direction of NASA with a second selected proposal (PI Michael Delp, PhD ; Radiation, Simulated Weightlessness and Countermeasures: Effects on Cerebral and Coronary Vascular Function and Structure). This descoped application is designed to address key aspects of the research goals put forward in Topic 6 of NRA NNJ16ZSA001N-SRHHC “Space Radiation and Human Health Countermeasures Topics”: 1) to provide detailed physiological and bio-molecular characterization of risk and prediction for cardiac disease from microgravity and space-like radiation doses that are mission relevant for future human spaceflight beyond low Earth orbit and 2) to determine relative biological effectiveness (RBE) values for space radiation in appropriate animals using relevant intermediate as well as late physiological endpoints and effects. These experimentally derived RBE values will provide quantitative inputs into methods and models to calculate degenerative tissue responses in humans following exposures in space. The proposed studies center on assessment of the degenerative tissue effects of space radiation on cardiac disease. The central hypothesis of this project is that simulated weightlessness interacts with whole body exposure to space relevant doses of radiation to potentiate risk for cardiac disease. Three specific aims are proposed to directly address the issue of risk for cardiac disease using experimental approaches in a rat model relevant to the research questions. The proposed investigations directly address important aspects of the NASA Human Research Program (HRP) Degenerative Tissue Gaps, as set forth in the Human Research Roadmap. HRP gaps are identified and addressed by each Specific Aim. The proposed research will provide important information to help close these gaps. Specific Aim A: Simulate a 3-3.5 year deep space human mission in male WAG rats, determining the impact of simulated weightlessness combined with components of galactic cosmic rays using a simplified 5-ion mixed field beam with total doses of 0.75 Gy (provisionally) and 1.50 Gy plus appropriate controls on risk for and occurrence of cardiac disease. Specific Aim B: Determine the response to gamma radiation in male Fischer 344 rats with and without simulated weightlessness to estimate relative biological effectiveness (RBE) for early clinically relevant endpoints of risk for and occurrence of cardiac disease. Rats will be irradiated with 3 doses of 137Cs: 3.0 -5.0 Gy (exact value to be determined), 8.0 Gy, and 10.0 Gy at Brookhaven National Laboratory alongside experimental animals. Specific Aim C: Simulate a 3-3.5 year deep space human mission in male Fischer 344 rats, determining the impact of simulated weightlessness combined with components of galactic cosmic rays using a simplified 5-ion mixed field beam with total doses of 0.75 Gy (provisionally) and 1.50 Gy plus appropriate controls on risk for and occurrence of cardiac disease.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research benefits life on Earth as it relates to patients who receive therapeutic radiation for the treatment of cancer and to individuals who receive accidental exposure to radiation from a nuclear accident.
Task Progress:	Exploration missions will take astronauts beyond the protection of the Earth’s magnetic field, where they will be continuously exposed to low levels of galactic cosmic rays (GCR) as well as microgravity. In addition to the increased risk for developing cancers, recent evidence suggests that exposure to heavy ion radiation that is part of the GCR spectrum may also accelerate the development of cardiovascular disease (CVD). The majority of the dose to astronauts from the GCR behind shielding is comprised of sparsely ionizing protons with low linear energy transfer, while the heavier ions are densely ionizing with high linear energy transfer. Our recent ground-based studies irradiated the total body of male WAG/RijCmcr (Wistar) rats, 6 months of age, with sequential exposures to rapidly switched beams of sparsely ionizing protons (1000 MeV, 80%), and more densely ionizing 28Si ions (500 MeV/n, 10%) and 56Fe ions (600 MeV/n, 10%) with total doses of 0.25, 0.50, 0.75, or 1.5 Gy. A risk factor for cardiac disease (serum total cholesterol) was modestly increased up to 9 months after total body irradiation of male Wistar rats after low to moderate doses (0.25 - 0.75 Gy) of this “mixed“ field exposure. Of note, increases in both systolic blood pressure and cardiac perivascular fibrosis were present 9 months after the highest dose (1.5 Gy) radiation. There were no changes in blood pressure or perivascular collagen deposition after 0.25, 0.50, or 0.75 Gy. Biomarkers of oxidative stress and inflammation increased 1 month after irradiation and decreased 2 months after irradiation with mixed beams. We determined the response of male Wistar rats at 6 months of age to sparsely ionizing gamma rays (1.5, 3.0, or 6.0 Gy) using 137Cs, and at 6 months or 6 weeks of age to X-rays (6.0, 8.0, or 10.0 Gy). Total cholesterol was unchanged in gamma irradiated rats. Total cholesterol was increased after 10.0 Gy of X-rays in 6 month and 6 week old rats compared with age-matched sham-irradiated controls. Cardiac perivascular collagen content was significantly increased in rats that were irradiated with 8.0 Gy of X-rays at 6 weeks of age but not at 6 months of age. While systemic blood pressure was elevated in both cohorts after 8.0 Gy X-ray irradiation (compared to age-matched sham-irradiated controls), the magnitude of the increase above baseline was greater in the younger rats. These findings indicate that dose and age at time of irradiation determine the timeline and severity of cardiac and renal injury after photon irradiation. Determination of relative biological effectiveness for GCRs using gamma rays as the reference radiation will need rats irradiated with higher doses of gamma together with a significant biological response from the two radiation types. The interaction between simulated weightlessness and space radiation on CVD needs to be understood prior to undertaking deep space exploratory missions. To date, combining the simulated spaceflight hazards of reduced weight bearing via hind limb unloading in rats of any strain with simultaneous GCR or gamma exposure in conscious animals has not been possible to conduct at the NASA Space Radiation Laboratory. To meet this challenge, we have engineered a solution that provides a realistic scenario for the investigation of exploratory spaceflight. Thus, our current ground-based studies directly address this important aspect of the NASA Human Research Program Degenerative Tissue Gaps as set forth in the Human Research Roadmap. The proposed research will provide important information to help close these gaps. We are determining the combined effects of simulated weightlessness and radiation exposure during deep space exploration on risk for and occurrence of CVD in a ground-based rat model. These studies simulate a 3-3.5 year deep space human mission in male Wistar rats 8-9 months old at the start of the study, determining the impact of simulated weightlessness combined with components of GCRs using a simplified 5-ion mixed field beam with total doses of 0.75

	Gy and 1.50 Gy plus appropriate controls on risk for CVD. Parallel studies are also underway with male rats from a second strain (Fischer 344) at an age representative of the astronaut corps.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Lenarczyk M, Kronenberg A, Mader M, North PE, Komorowski R, Cheng Q, Little MP, Chiang IH, LaTessa C, Jardine J, Baker JE. "Age at exposure to radiation determines severity of renal and cardiac disease in rats." Radiat Res. 2019 Jul;192(1):63-74. Epub 2019 May 16. https://doi.org/10.1667/RR15043.1 ; PubMed PMID: 31095446 , Jul-2019